Muhammad Hamza

AI/ML Engineer | Generative AI, Agentic Systems & RAG

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AI Engineer specializing in agentic AI systems, LLM application development, and Retrieval-Augmented Generation (RAG). Hands-on experience designing multi-agent workflows with LangChain and LangGraph, building production RAG pipelines over vector databases (FAISS, ChromaDB), and shipping LLM-powered automation for enterprise, cybersecurity, and backend systems. Background spans end-to-end AI development: data preprocessing, model training and evaluation, prompt engineering, API integration, and deployment, with published research in 3D deep learning for Alzheimer's disease detection from MRI.

CORE TECHNICAL SKILLS

Agentic AI, Generative AI & LLM Orchestration
LangChain, LangGraph (multi-agent workflows, state machines, tool use), Model Context Protocol (MCP), Large Language Models (LLMs), Prompt Engineering, LLM Application Development

RAG, Vector Search & Information Retrieval
RAG Pipeline Architecture, LlamaIndex, Haystack, Vector Search & Databases (FAISS, ChromaDB), Embedding Generation, Semantic Search, Document Chunking & Indexing

Machine Learning & Deep Learning
Machine Learning (ML), Deep Learning (DL), Natural Language Processing (NLP), Model Training & Evaluation, Data Preprocessing, Deep Learning Architectures (CNN, TabNet, Transformers)

Programming, Cloud & DevOps
Python, C++, NumPy, Pandas, Matplotlib, Seaborn, Jupyter Notebook, VS Code, GitHub, AWS, GitHub Actions, CI/CD, Jenkins, Docker, Kubernetes, Postman

EXPERIENCE

CyberZeus Software Systems, *AI Developer* — 01/2026 – Present, Lahore, Pakistan

- Designed and developed AI/ML solutions for enterprise applications, applying agentic AI design patterns to improve workflows.
- Owned the architecture and end-to-end implementation of agentic AI and RAG systems, from workflow design and retrieval strategy to deployment.
- Led technical implementation decisions for LangGraph-based multi-agent workflows, including agent state management, tool orchestration, context handling, and FAISS/ChromaDB-based retrieval for automated technical analysis.
- Guided developers on AI/ML implementation approaches, reviewed integrations, and helped troubleshoot model, RAG, and backend issues across active projects.
- Developed AI-driven cybersecurity systems for anomaly detection and intelligent threat analysis, with evaluation based on accuracy, precision, recall & F1-score.

Efaida Software House, *AI Software Engineer* — 08/2025 – 12/2025, Okara, Pakistan

- Developed and automated AI-driven workflows combining Python-based ML pipelines with Zapier, n8n, and Make.com, improving reporting speed and operational productivity.
- Developed AI models for text classification and object detection, integrating model outputs into automated downstream workflows.
- Used Git-based version control and CI/CD workflows (GitHub Actions) to test and deploy automation scripts reliably.

Minhaj Solutions, *AI Developer (Part-Time)* — 04/2024 – 08/2025, Renala Khurd, Pakistan

- Built an agentic RAG-based support assistant that used LLMs to understand and resolve technical product queries.
- Replaced traditional rule-based responses with context-aware answers generated from relevant product knowledge.
- Developed the embedding, semantic search, and response-generation pipeline to provide real-time, knowledge-grounded support.
- Worked on applied AI and machine learning projects using real-world medical imaging and classification datasets.

Fiverr, *Freelance Python Developer & ML Engineer* — 07/2023 – Present, Remote

- Completed 12+ international projects in Python automation, machine learning, and data analysis as a Level 1 Fiverr Seller.
- Delivered end-to-end ML solutions including preprocessing, model training, evaluation, deployment, and reporting workflows.

PROJECTS

Agentic Cybersecurity AI Assistant — Multi-Agent RAG System

Enterprise-grade AI cybersecurity assistant built with LangChain and LangGraph, combining multi-agent orchestration, Retrieval-Augmented Generation (RAG), vector search, external security tools, and automated reporting to support vulnerability analysis and continuous monitoring.

- Architected multi-agent workflows with specialized planner, retriever, and analyst agents for automated vulnerability analysis, reasoning, and report generation.
- Engineered the RAG pipeline using FAISS/ChromaDB, embeddings, semantic search, and vulnerability knowledge bases to provide context-grounded security analysis.
- Designed fallback/error-handling strategies for failed tool calls and incomplete retrieval results.
- Optimized token utilization and inference latency through context-window management, selective retrieval, and efficient prompt construction.
- Implemented persistent session memory and contextual state management to maintain continuity across multi-step agent workflows.
- Implemented CI/CD pipelines using GitHub Actions and containerized agent services with Docker.

AI-Based Ticket Scalping Prevention System

- Developed an AI-based fraud detection and dynamic pricing system for ticketing platforms
- Built anomaly detection models to identify bot activity and suspicious transactions
- Implemented pricing and fraud prevention mechanisms for fair ticket distribution
- Implemented CI/CD workflows via GitHub Actions to automate model retraining, testing, and deployment to production.

AI-Driven Intrusion Detection for Critical Systems

- Designed a hybrid deep-learning intrusion detection architecture combining TabNet and Transformers
- Trained and evaluated models on the CIC-IDS 2017 cybersecurity dataset
- Applied attention-based feature learning to improve classification of malicious network-traffic patterns.
- Deployed the detection model using Docker and Kubernetes for scalable, real-time monitoring across critical infrastructure nodes.

Agentic RAG-Powered Support Assistant

- Designed an agentic RAG-powered support assistant that autonomously resolves technical product queries using LLM-driven reasoning
- Implemented retrieval-augmented, context-aware response generation in place of rule-based query
- Built embedding, semantic search, and automated response-generation pipelines powering real-time, knowledge-grounded conversations

EDUCATION

University of Okara, *BS Software Engineering* — 2020 – 2024

Punjab Group of Colleges, *FSC* — 2017 – 2019

CERTIFICATIONS & ACHIEVEMENTS

Fully Funded Scholar — Design & Development of Unmanned Aerial Systems

Samara National Research University, Russia | Nomination by the University of Okara | 2025

Kaggle Notebook Expert Badge, Kaggle

Ranked Top 2,100 of ~60,000+ contributors in Kaggle Notebooks.

RESEARCH & PUBLICATIONS

DOI: https://doi.org/10.1016/j.asej.2025.103714

Created a 3D deep learning model (MobiBrainNet) for multi-class Alzheimer's detection using volumetric MRI data. The work enhanced classification accuracy while focusing on computational efficiency and spatial data modeling.